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# ADRIAIndicators.jl: a Julia package for summarizing reef ecological model outputs

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## Summary

ADRIAIndicators.jl is a Julia package for analyzing outputs from coral reef ecological models. Its primary purpose is to provide a standardized and dependency-free toolkit for transforming high-dimensional model outputs such as coral abundance by reef, time, species, and size class into lower-dimensional, interpretable metrics. The package offers a wide range of functions, from simple aggregations and unit conversions to more complex indices and estimators derived from regression models. These tools help with the estimation of functional diversity, juvenile abundance, shelter volume, fish biomass, and overall reef condition, enabling consistent and comparable analysis across different coral ecology models such as CoralBlox (Ribeiro de Almeida et al., 2025, 2026), C~Scape (Cresswell et al., 2024), ReefMod (Bozec et al., 2022), and CoCoNet (Condie, 2022).

## Statement of Need

Models of coral reef ecosystems often produce large volumes of high-dimensional data. There is a need for standardized tools to summarize and analyze these model outputs to facilitate inter-model comparison of environmental projections and communicate results to managers and stakeholders. ADRIAIndicators.jl provides a set of standard indicator metrics that can be used to summarize reef state in ecological model outputs. These were originally implemented within the ADRIA.jl Decision Support package (Iwanaga et al., 2025) but were being reproduced in many workflows that did not use ADRIA.jl. Separating these indicators from ADRIA.jl is part of efforts to better compartmentalize code in a more reusable and interoperable manner.

ADRIAIndicators.jl is written in Julia (Bezanson et al., 2017), a high-level, high-performance programming language for technical computing. This package is designed to be easy to use, and provides an in-place option for all metrics for any eventual wrappers that may be implemented in other languages such as Python and R. Such wrappers could be developed leveraging Julia's support for language interoperability and compilation capabilities (such as those provided by JuliaC.jl).

## State of the Field

In the field of coral reef ecology, models such as ReefMod (Bozec et al., 2022), C~Scape (Cresswell et al., 2024), and CoCoNet (Condie, 2022) have historically employed internal, custom-coded implementations for calculating ecological indicators. This lack of modularity often leads to duplicate development efforts for common metrics (e.g., Simpson's Diversity or unit conversions) and can introduce subtle inconsistencies in the calculation of more complex indicators like Shelter Volume (Urbina-Barreto et al., 2021) or the Reef Condition Index (Heneghan et al., 2026).

ADRIAIndicators.jl addresses these challenges by providing a standalone, model-agnostic library that decouples ecological indicators from specific ecological models. This approach allows researchers to apply identical analytical pipelines to outputs from diverse modeling platforms, facilitating rigorous cross-model benchmarking and ensuring that management decisions are based on comparable ecological signals.

# Software Design

ADRIAIndicators.jl is designed for high performance and minimal overhead, leveraging Julia's (Bezanson et al., 2017) strengths in numerical computing. The package uses multiple dispatch to handle different output dimensions (e.g., 4D arrays for single scenarios vs. 5D arrays for ensemble outputs) while maintaining a consistent API. Key design features include:

- **In-place Operations:** All metrics provide mutating versions (indicated by appending `!` to the function name) to allow for memory-efficient processing of large-scale reef simulation data without unnecessary allocations.
- **Dependency-Free:** The core library has zero external package dependencies, ensuring long-term stability and ease of integration into other modeling suites.
- **Dimensional Consistency:** Functions strictly adhere to the `[Time, Groups, Sizes, Locations, Scenarios]` dimensional order, simplifying data preparation for end-users.

## Available Indicators

The indicators implemented in ADRIAIndicators.jl are classified into three categories: Aggregations, Conversions, and Metrics. Aggregations are convenience methods for reducing the dimensionality of data by summarizing arrays. Conversions handle transformations between different units or representations of coral cover. Metrics derive higher-level, interpretable indicators from the raw model data, such as coral diversity, shelter volume, and composite indices for reef health.

| Metric Name | Type | Reference |
|---|---|---|
| Relative Cover | Aggregation | |
| Relative Location Taxonomy Cover | Aggregation | |
| Relative Taxonomy Cover | Aggregation | |
| LTMP Cover | Aggregation | |
| LTMP Location Taxonomy Cover | Aggregation | |
| LTMP Taxonomy Cover | Aggregation | |
| Relative Juveniles | Aggregation | |
| Relative Location Taxonomy Juveniles | Aggregation | |
| Relative Taxonomy Juveniles | Aggregation | |
| Relative Habitable Cover to Reef Cover | Conversion | |
| Reef Cover to Relative Habitable Cover | Conversion | |
| Absolute Shelter Volume | Metric | (Aston et al., 2022; Urbina-Barreto et al., 2021) |
| Relative Shelter Volume | Metric | - |
| Coral Diversity | Metric | (Hill, 1973) |
| Coral Evenness | Metric | - |

| Metric Name | Type | Reference |
|---|---|---|
| Reef Condition Index | Metric | (Heneghan et al., 2026) |
| Reef Fish Index | Metric | (Graham & Nash, 2013) |

*A dash (-) in the 'Reference' column indicates the reference is the same as the entry directly above it.*

## Usage

The order of dimensions is always the same in ADRIAIndicators.jl,

1. Time
2. Groups
3. Sizes
4. Locations
5. Scenarios

If a dimension is missing then the order remains the same however the missing dimensions are excluded. Furthermore, all metrics have an option to provide a buffer as input in the cases where one wants to write the metric into an existing array or sub-array. This implementation is relied upon by functions that allocate the returned array as-well and was chosen to account for any decisions in the future where another language may wrap this library and need to pass memory that is not managed by Julia.

```julia
using ADRIAIndicators

# Create some dummy model output data
n_timesteps = 75
n_groups = 5
n_sizes = 7
n_locations = 3806

# Raw model coral cover outputs with dimensions [timesteps · groups · sizes · locations]
raw_model_cover = rand(Float64, n_timesteps, n_groups, n_sizes, n_locations);

# Juveniles mask with dimensions [sizes]
is_juvenile = [true, true, false, false, false, false, false];

# Calculate and allocate new array for metric
rel_juveniles = relative_juveniles(raw_model_cover, is_juvenile);

# Users can provide output buffers if it is more convenient for them
rel_juveniles_out = zeros(Float64, n_timesteps, n_locations);
relative_juveniles!(raw_model_cover, is_juvenile, rel_juveniles_out);
```

# Research Impact

ADRIAIndicators.jl provides a standardized collection of metrics for the analysis of reef model outputs, supporting consistent comparative studies and ensemble modeling efforts. By establishing a shared library of indicators, the package ensures that different simulation platforms can summarize raw outputs using identical mathematical formulations, facilitating like-for-like comparison across workflows.

Currently, the package provides the mathematical definitions and computations for the metrics module of ADRIA.jl (Iwanaga et al., 2025), a decision-support platform used to evaluate coral reef intervention decisions. In this workflow, ADRIAIndicators.jl summarizes the high-dimensional spatial-demographic projections from CoralBlox (Ribeiro de Almeida et al., 2025) into standardized indicators, such as relative shelter volume and reef condition indices, which are then fed directly into ADRIA's decision-support algorithms. Similarly, CScapeInterface.jl (Haller-Bull, 2026) leverages the package to post-process and export C~Scape (Cresswell et al., 2024) simulation results, allowing these high-fidelity spatial projections to also be fed into ADRIA's analysis platform for the same analysis as CoralBlox.jl. Furthermore, usage is planned for the ReefGuide tool (White et al., 2026), a web-based assessment platform.

Decoupling these indicators from ecological models ensures that different coral ecology models such as CoralBlox, C~Scape, or observational data are summarized consistently. This standardized definition of metrics is essential for large-scale collaborative initiatives like the Reef Restoration and Adaptation Program (RRAP), while the package's zero-allocation design ensures that post-processing massive simulation ensembles does not become a computational bottleneck. Additionally, the package facilitates the sharing of metrics with other reef systems beyond the Great Barrier Reef, where they are currently used.

# AI Usage Disclosure

Gemini 2.5 Pro and Gemini 3 Pro were used in the development of tests for all implemented metrics in ADRIAIndicators.jl and were reviewed by the authors to ensure correctness.

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